

Challenge: Kangaroo Rats, Diets, and Body Size

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Set-up

1. Import the Data

```
[1] "1: dimensions (rows, columns) of each data frame"
```

```
[1] 35549      9
```

```
[1] 24      2
```

```
[1] 54      4
```

```
[1] 5400     26
```

2. Tidy the Trait Table

```
[1] "2"
```

```
# A tibble: 5,400 x 12
```

	sci_name	mass_g	`Diet-Inv`	`Diet-Vend`	`Diet-Vect`	`Diet-Vfish`	`Diet-Vunk`
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Tachyglos~	3025	100	0	0	0	0
2	Zaglossus~	8532.	100	0	0	0	0
3	Zaglossus~	7180	100	0	0	0	0
4	Zaglossus~	10140.	100	0	0	0	0

```

5 Ornithorh~ 1484.      80      0      0      20      0
6 Caluromys~ 229.      20      0      0      0      10
7 Caluromys~ 297      20      0      0      0      10
8 Caluromys~ 325      20      0      0      0      10
9 Caluromys~ 250      20      0      0      0      0
10 Glironia ~ 114      40      0      0      0      20
# i 5,390 more rows
# i 5 more variables: `Diet-Scav` <dbl>, `Diet-Fruit` <dbl>, `Diet-Nect` <dbl>,
#   `Diet-Seed` <dbl>, `Diet-Plant0` <dbl>

```

```
[1] 5400 12
```

3. Match Portal Species to Traits

```
[1] "3a"
```

```

# A tibble: 31 x 5
  species_id sci_name      genus      species      taxa
  <chr>      <chr>      <chr>      <chr>      <chr>
1 AH      Ammospermophilus harrisi Ammospermophilus harrisi Rodent
2 BA      Baiomys taylori      Baiomys      taylori Rodent
3 DM      Dipodomys merriami      Dipodomys      merriami Rodent
4 DO      Dipodomys ordii      Dipodomys      ordii Rodent
5 DS      Dipodomys spectabilis      Dipodomys      spectabilis Rodent
6 DX      Dipodomys sp.      Dipodomys      sp. Rodent
7 NL      Neotoma albigula      Neotoma      albigula Rodent
8 NX      Neotoma sp.      Neotoma      sp. Rodent
9 OL      Onychomys leucogaster      Onychomys      leucogaster Rodent
10 OT      Onychomys torridus      Onychomys      torridus Rodent
# i 21 more rows

```

```
[1] "3b: rodent species with no match in the trait table"
```

```

# A tibble: 9 x 5
  species_id sci_name      genus      species      mass_g
  <chr>      <chr>      <chr>      <chr>      <dbl>
1 AH      Ammospermophilus harrisi Ammospermophilus harrisi      NA
2 DX      Dipodomys sp.      Dipodomys      sp.      NA
3 NX      Neotoma sp.      Neotoma      sp.      NA
4 OX      Onychomys sp.      Onychomys      sp.      NA
5 PH      Perognathus hispidus      Perognathus      hispidus      NA

```

6 PX	Chaetodipus sp.	Chaetodipus	sp.	NA
7 RX	Reithrodontomys sp.	Reithrodontomys	sp.	NA
8 SX	Sigmodon sp.	Sigmodon	sp.	NA
9 UR	Rodent sp.	Rodent	sp.	NA

[1] "3c: 24 species"

A tibble: 24 x 16

	species_id	sci_name	genus	species	taxa	mass_g	`Diet-Inv`	`Diet-Vend`
	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1 AH	Ammospermophilu~	Ammo~	harris~	Rode~		122	20	0
2 BA	Baiomys taylori	Baio~	taylori	Rode~		8	10	0
3 DM	Dipodomys merri~	Dipo~	merria~	Rode~		42	20	0
4 DO	Dipodomys ordii	Dipo~	ordii	Rode~		60.4	20	0
5 DS	Dipodomys spect~	Dipo~	specta~	Rode~		136.	20	0
6 NL	Neotoma albigula	Neot~	albigu~	Rode~		206.	10	0
7 OL	Onychomys leuco~	Onyc~	leucog~	Rode~		27.9	90	0
8 OT	Onychomys torri~	Onyc~	torrid~	Rode~		25.0	90	0
9 PB	Chaetodipus bai~	Chae~	baileyi	Rode~		26.4	10	0
10 PE	Peromyscus erem~	Pero~	eremic~	Rode~		23.6	50	0

i 14 more rows

i 8 more variables: `Diet-Vect` <dbl>, `Diet-Vfish` <dbl>, `Diet-Vunk` <dbl>,
 # `Diet-Scav` <dbl>, `Diet-Fruit` <dbl>, `Diet-Nect` <dbl>,
 # `Diet-Seed` <dbl>, `Diet-Plant0` <dbl>

[1] 24

4. Build the Analysis Table

[1] "4"

A tibble: 20,451 x 25

	record_id	month	day	year	plot_id	species_id	sex	hindfoot_length	weight
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>
1	1	7	16	1977	2	NL	M	32	NA
2	2	7	16	1977	3	NL	M	33	NA
3	3	7	16	1977	2	DM	F	37	NA
4	5	7	16	1977	3	DM	M	35	NA
5	7	7	16	1977	2	PE	F	NA	NA
6	13	7	16	1977	3	DM	M	35	NA

```

7      14      7      16 1977      8 DM      <NA>      NA      NA
8      16      7      16 1977      4 DM      F      36      NA
9      17      7      16 1977      3 DS      F      48      NA
10     18      7      16 1977      2 PP      M      22      NA

```

```
# i 20,441 more rows
```

```
# i 16 more variables: sci_name <chr>, genus <chr>, species <chr>, taxa <chr>,
#   mass_g <dbl>, `Diet-Inv` <dbl>, `Diet-Vend` <dbl>, `Diet-Vect` <dbl>,
#   `Diet-Vfish` <dbl>, `Diet-Vunk` <dbl>, `Diet-Scav` <dbl>,
#   `Diet-Fruit` <dbl>, `Diet-Nect` <dbl>, `Diet-Seed` <dbl>,
#   `Diet-Plant0` <dbl>, plot_type <chr>
```

```
[1] "4: captures per plot type"
```

```
# A tibble: 2 x 2
```

	plot_type <chr>	n_captures <int>
1	Control	15428
2	Long-term Krat Exclosure	5023

5. Relative Abundance of Kangaroo Rats

```
[1] "5a"
```

```
# A tibble: 46 x 5
```

	plot_type <chr>	genus <chr>	species <chr>	n <int>	rel_abund <dbl>
1	Control	Dipodomys	merriami	6760	0.438
2	Control	Dipodomys	ordii	1671	0.108
3	Control	Dipodomys	spectabilis	1389	0.0900
4	Control	Chaetodipus	penicillatus	1108	0.0718
5	Control	Onychomys	torridus	1004	0.0651
6	Control	Chaetodipus	baileyi	789	0.0511
7	Control	Neotoma	albigula	610	0.0395
8	Control	Onychomys	leucogaster	471	0.0305
9	Control	Perognathus	flavus	424	0.0275
10	Control	Reithrodontomys	megalotis	403	0.0261

```
# i 36 more rows
```

```
[1] "5a: relative abundances sum to 1 within each plot type"
```

```
# A tibble: 2 x 2
  plot_type      total
  <chr>         <dbl>
1 Control             1
2 Long-term Krat Exclosure 1
```

```
[1] "5b"
```

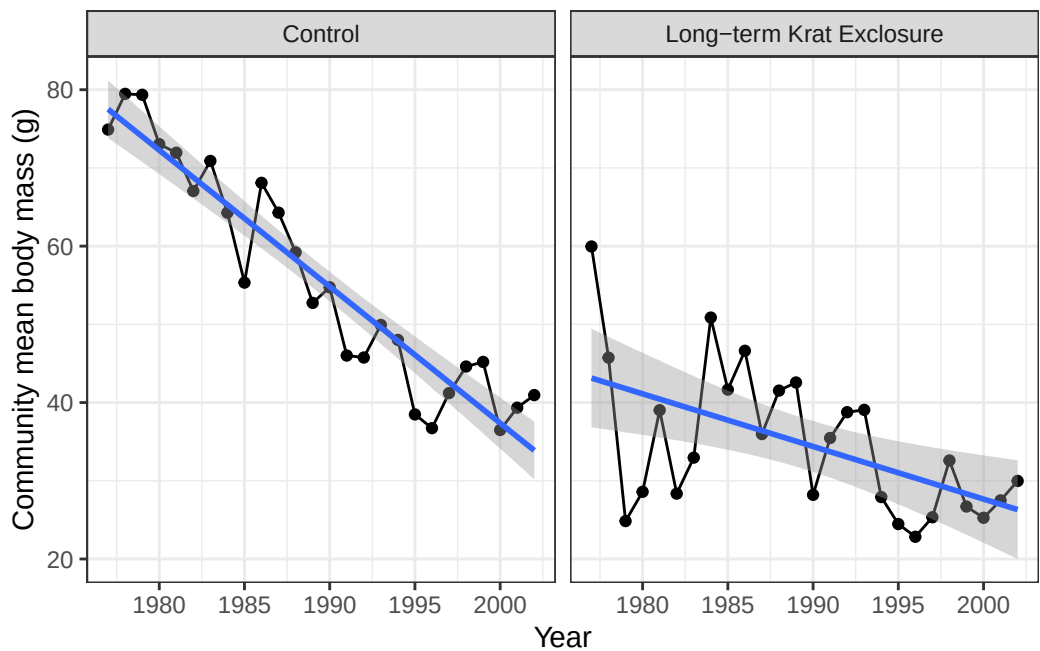
```
# A tibble: 2 x 2
  plot_type      prop_krats
  <chr>         <dbl>
1 Control             0.637
2 Long-term Krat Exclosure 0.0709
```

6. Body Size Through Time

```
[1] "6a"
```

```
# A tibble: 52 x 4
  year plot_type      n_captures mean_mass_g
  <dbl> <chr>         <int>         <dbl>
1  1977 Control             221           74.9
2  1977 Long-term Krat Exclosure 61           60.0
3  1978 Control             500           79.5
4  1978 Long-term Krat Exclosure 78           45.7
5  1979 Control             321           79.3
6  1979 Long-term Krat Exclosure 62           24.8
7  1980 Control             612           73.0
8  1980 Long-term Krat Exclosure 151           28.6
9  1981 Control             644           71.9
10 1981 Long-term Krat Exclosure 103           39.0
# i 42 more rows
```

```
[1] "6b"
```



7. Community Diet

[1] "7a"

A tibble: 204,510 x 5

	record_id	sci_name	plot_type	diet	diet_percent
	<dbl>	<chr>	<chr>	<chr>	<dbl>
1	1	Neotoma albigula	Control	Inv	10
2	1	Neotoma albigula	Control	Vend	0
3	1	Neotoma albigula	Control	Vect	0
4	1	Neotoma albigula	Control	Vfish	0
5	1	Neotoma albigula	Control	Vunk	0
6	1	Neotoma albigula	Control	Scav	0
7	1	Neotoma albigula	Control	Fruit	0
8	1	Neotoma albigula	Control	Nect	0
9	1	Neotoma albigula	Control	Seed	20
10	1	Neotoma albigula	Control	Plant0	70

i 204,500 more rows

[1] "7b"

```
# A tibble: 20 x 3
  plot_type      diet mean_percent
  <chr>         <chr>         <dbl>
1 Control      Fruit          13.5
2 Control      Inv           25.3
3 Control      Nect            0
4 Control      Plant0         26.4
5 Control      Scav           0.382
6 Control      Seed           34.4
7 Control      Vect            0
8 Control      Vend            0
9 Control      Vfish           0
10 Control     Vunk            0
11 Long-term Krat Exclosure Fruit          3.72
12 Long-term Krat Exclosure Inv           21.8
13 Long-term Krat Exclosure Nect            0
14 Long-term Krat Exclosure Plant0         20.4
15 Long-term Krat Exclosure Scav            1.15
16 Long-term Krat Exclosure Seed           52.9
17 Long-term Krat Exclosure Vect            0
18 Long-term Krat Exclosure Vend            0
19 Long-term Krat Exclosure Vfish           0
20 Long-term Krat Exclosure Vunk            0
```

```
[1] "7b: each plot type sums to 100"
```

```
# A tibble: 2 x 2
  plot_type      total
  <chr>         <dbl>
1 Control      100.
2 Long-term Krat Exclosure 100.0
```

```
[1] "7c"
```

```
# A tibble: 10 x 3
  plot_type      diet mean_percent
  <chr>         <chr>         <dbl>
1 Control      Fruit          13.5
2 Control      Inv           25.3
3 Control      Plant0         26.4
4 Control      Scav           0.382
```

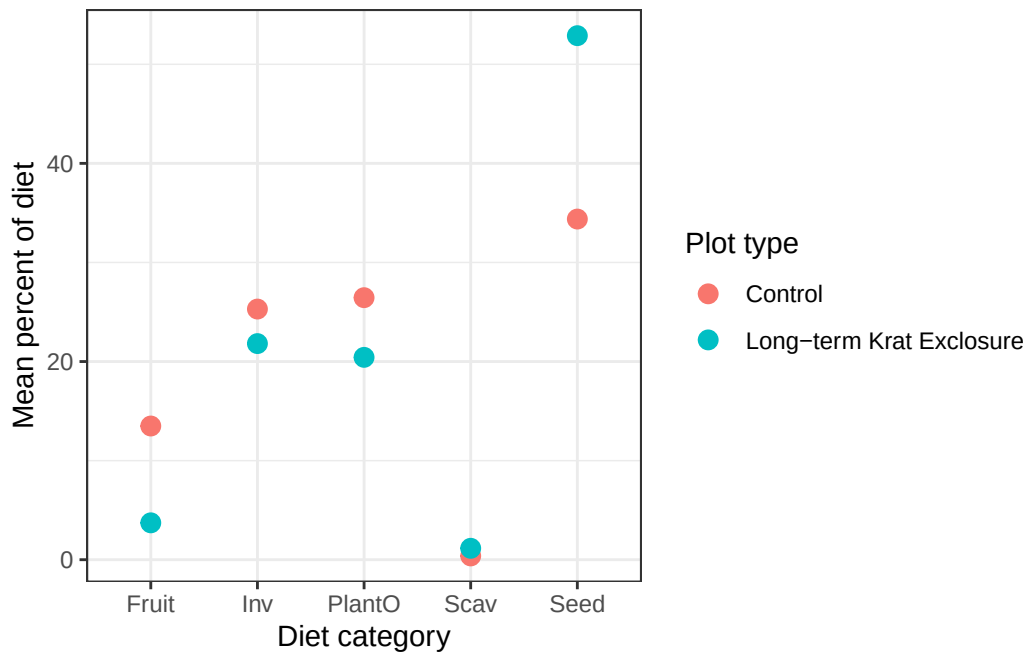
5	Control	Seed	34.4
6	Long-term Krat Exclosure	Fruit	3.72
7	Long-term Krat Exclosure	Inv	21.8
8	Long-term Krat Exclosure	PlantO	20.4
9	Long-term Krat Exclosure	Scav	1.15
10	Long-term Krat Exclosure	Seed	52.9

[1] "7d"

A tibble: 5 x 4

	diet	control	exclosure	difference
	<chr>	<dbl>	<dbl>	<dbl>
1	Seed	34.4	52.9	18.5
2	Scav	0.382	1.15	0.771
3	Inv	25.3	21.8	-3.49
4	PlantO	26.4	20.4	-6.02
5	Fruit	13.5	3.72	-9.77

[1] "7e"



Extension: Do the Database Masses Match What Portal Measured?

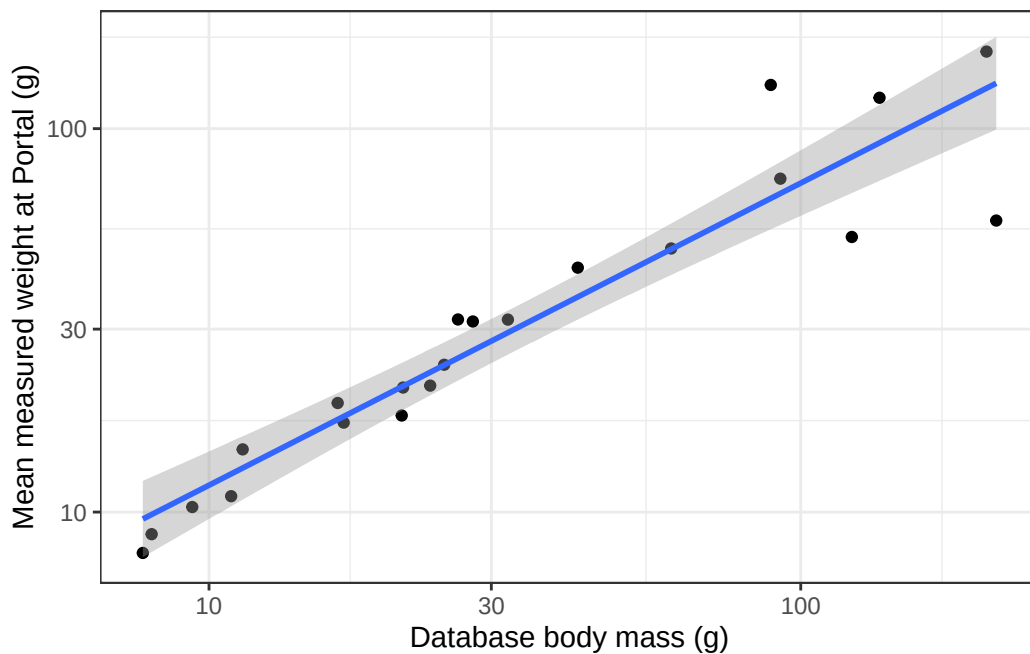
```
[1] "Extension a-b"
```

```
# A tibble: 22 x 4
```

	sci_name <chr>	mass_g <dbl>	n_captures <int>	mean_weight_g <dbl>
1	Baiomys taylori	8	38	8.76
2	Chaetodipus baileyi	26.4	1836	31.8
3	Chaetodipus hispidus	32	22	31.8
4	Chaetodipus intermedius	16.5	9	19.2
5	Chaetodipus penicillatus	16.9	1987	17.1
6	Dipodomys merriami	42	7005	43.4
7	Dipodomys ordii	60.4	1756	48.7
8	Dipodomys spectabilis	136.	1415	120.
9	Neotoma albigula	206.	776	159.
10	Onychomys leucogaster	27.9	579	31.4

```
# i 12 more rows
```

```
[1] "Extension c"
```



```
[1] "Extension d"
```

```

# A tibble: 22 x 5
  sci_name          n_captures mass_g mean_weight_g ratio
  <chr>          <int>    <dbl>      <dbl> <dbl>
1 Sigmodon fulviventer      38    214         57.6  3.72
2 Sigmodon ochrognathus     31    122         52.2  2.34
3 Neotoma albigula       776   206.         159.   1.30
4 Sigmodon hispidus       119    92.4         74.0  1.25
5 Dipodomys ordii      1756    60.4         48.7  1.24
6 Peromyscus leucopus        7    21.2         17.9  1.19
7 Dipodomys spectabilis   1415   136.         120.   1.13
8 Peromyscus eremicus     557    23.6         21.4  1.11
9 Onychomys torridus     1381    25.0         24.2  1.03
10 Chaetodipus hispidus     22     32         31.8  1.01
# i 12 more rows

```